

BAENGYEONGDO AQMS, Korea, Republic of

WMO: 471030

Lat: 37.966N

Lon: 124.630E

Elev: 126

StdP: 99.82

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.2	(c) -7.1	(d) -12.8	(e) 1.3	(f) -6.7	(g) -11.3	(h) 1.4	(i) -6.0	(j) 13.1	(k) -2.1	(l) 11.5	(m) -2.2	(n) 4.6	(o) 30	(p) 0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.3	(c) 29.2	(d) 25.1	(e) 28.0	(f) 24.3	(g) 26.8	(h) 23.6	(i) 26.1	(j) 28.2	(k) 25.3	(l) 27.0	(m) 24.6	(n) 26.1	(o) 2.3	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	21.0	27.2	24.9	20.2	26.4	24.0	19.2	25.6	81.9	29.1	77.8	26.5	74.5	26.1	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.9	(b) 9.0	(c) 7.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 10.9	(5) 9.0	(6) 7.7	(7) -11.0	(8) 27.0	(9) 1.6	(10) 1.1	(11) -12.2	(12) 27.8	(13) -13.1	(14) 28.5	(15) -14.0	(16) 29.1	(17) -15.1	(18) 30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-1.3	-0.1	3.8	9.3	14.5	19.4	23.0	24.1	20.2	14.7	7.7	2.3
	DBStd	9.21	3.40	3.46	2.78	3.35	2.79	2.42	2.32	2.16	2.33	3.12	4.09	3.50
	HDD10.0	1211	351	283	192	51	1	0	0	0	0	5	88	239
	HDD18.3	2928	609	517	451	271	123	17	0	9	116	318	497	
	CDD10.0	1768	0	0	0	31	140	282	403	436	307	150	20	0
	CDD18.3	443	0	0	0	1	3	48	144	177	66	3	0	0
	CDH23.3	1781	0	0	0	2	8	104	608	938	118	2	0	0
	CDH26.7	296	0	0	0	1	0	10	115	167	4	0	0	0
Wind	WSAvg	3.4	4.0	3.4	3.9	4.1	3.6	2.9	3.0	2.9	2.7	3.0	3.7	3.7
Precipitation	PrecAvg	818	13	18	19	50	72	76	219	166	86	35	40	22
	PrecMax	1265	28	66	72	99	137	202	400	327	362	94	154	44
	PrecMin	406	2	0	2	6	3	16	74	36	6	1	10	7
	PrecStd	201	7	16	18	28	44	45	93	87	105	27	37	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	8.2	12.1	21.3	24.2	28.0	31.2	31.1	27.2	22.2	16.6	11.1
		MCWB	6.4	6.3	7.8	14.3	18.4	19.8	26.4	25.7	23.3	18.7	13.9	9.1
	2%	DB	6.0	6.9	10.1	18.0	21.8	25.8	29.2	29.8	25.9	20.9	15.2	9.4
		MCWB	4.8	5.0	6.9	12.6	16.0	19.6	25.3	25.4	22.6	17.5	12.4	7.9
	5%	DB	5.0	5.2	8.9	15.9	20.0	24.2	27.8	28.2	24.9	19.7	14.1	8.2
		MCWB	3.6	3.5	6.2	11.1	14.8	19.4	24.3	24.8	21.7	16.5	11.7	6.4
	10%	DB	3.8	4.2	7.7	14.1	18.8	23.0	26.2	27.1	23.8	18.2	13.0	7.1
		MCWB	2.4	2.6	5.2	9.9	14.2	18.6	23.7	24.1	20.9	15.4	10.8	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.6	8.7	16.3	18.7	22.2	27.4	27.3	25.2	19.7	15.2	9.9
		MCDB	7.3	8.1	10.2	19.5	23.1	24.6	29.9	29.4	26.2	21.4	16.0	10.6
	2%	WB	5.0	5.2	7.5	13.2	17.1	21.2	26.1	26.3	23.7	18.2	13.7	8.1
		MCDB	5.6	6.0	9.3	16.8	20.5	23.3	28.2	28.4	24.9	19.9	14.6	9.0
	5%	WB	3.8	4.1	6.4	11.9	15.9	20.6	25.2	25.6	22.5	17.2	12.2	6.8
		MCDB	4.7	5.0	8.2	15.1	18.6	22.6	26.9	27.5	24.1	18.9	13.6	7.9
	10%	WB	2.2	2.9	5.4	10.6	15.0	20.0	24.4	24.9	21.3	16.2	11.1	5.5
		MCDB	3.4	4.1	6.9	13.3	17.7	21.7	25.8	26.5	23.3	18.1	12.7	7.0
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.1	5.6	6.6	6.5	5.7	4.7	5.3	5.4	5.2	5.1	4.9
		MCDBR	6.2	5.6	6.9	9.2	8.4	7.7	6.7	6.4	6.2	6.0	5.7	5.6
		MCWBR	5.9	4.8	5.3	6.1	5.5	4.8	4.8	4.0	4.3	4.7	5.2	5.5
	5% WB	MCDBR	6.4	5.0	6.4	9.0	7.3	6.0	5.8	5.7	5.6	5.6	5.8	5.6
		MCWBR	6.5	4.9	5.4	6.6	5.5	4.3	4.4	3.9	4.4	5.2	6.0	5.7
Clear-Sky Solar Irradiance	taub	0.432	0.485	0.543	0.594	0.613	0.693	0.670	0.576	0.505	0.517	0.498	0.439	
	taud	1.961	1.860	1.760	1.702	1.716	1.627	1.725	1.941	2.038	1.883	1.878	1.948	
	Ebn at Noon	697	712	714	710	706	651	661	717	744	673	619	651	
	Edh at Noon	136	171	208	233	234	256	230	182	156	165	148	129	
All-Sky Solar Radiation	RadAvg	2.20	3.14	4.38	5.05	5.55	5.11	4.39	4.72	4.47	3.57	2.36	1.91	
	RadStd	0.17	0.23	0.36	0.35	0.51	0.62	0.59	0.57	0.38	0.25	0.28	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+84	+48

Nomenclature: See separate page